Sai Mukundhan Madhavan

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OBJECTIVE

M.Tech student specializing in Wireless Smart Communication and IoT, with hands-on experience in LoRaWAN systems, ESP32, STM32 microcontrollers, and cloud-connected dashboards. Currently interning at C-DAC Hyderabad, building smart IoT solutions. Seeking a challenging role in Embedded Systems or IoT Engineering where I can contribute to real-world product development.

INTERNSHIP

C-DAC (Centre for Development of Advanced Computing), Hyderabad — **Jan 2026 – Jun 2026 (Ongoing)**

- Developed an IoT-based Smart Parking System using STM32 Nucleo board and ToF (Time-of-Flight) sensors to measure real-time distance and detect vehicle occupancy in parking slots.
- Built SoilBot AI — a smart agriculture web application from scratch, featuring a full-stack implementation with HTML, CSS, and JavaScript (frontend) and Python Flask (backend), deployed live at soil-bot.vercel.app.
- Integrated an AI chatbot into SoilBot for real-time crop and soil health recommendations based on sensor data.
- Studied and explored V2X (Vehicle-to-Everything) Communication protocols and their applications in intelligent transportation systems.

EDUCATION

M.Tech – Wireless Smart Communication — 2024 – 2026
SASTRA Deemed University, Thanjavur

B.Tech – Electronics and Communication Engineering — 2019 – 2023
SRM Institute of Science and Technology, Chennai

Intermediate (Class XII) — 2017 – 2019
Raghava Laxmi Devi Government College, Hyderabad

TECHNICAL SKILLS

Programming	C, Python, MATLAB, JavaScript
Microcontrollers	Arduino, ESP32, STM32 Nucleo
Communication & IoT	LoRa SX1278/SX1262, LoRaWAN (TTN), GPS NEO-6M, V2X Communication, Sensor Integration
Web Development	HTML, CSS, JavaScript, Python Flask
Tools & Platforms	Arduino IDE, MATLAB, STM32CubeIDE, ThingSpeak, Vercel, TinyGPS++

AREAS OF INTEREST

- Internet of Things (IoT)
- Embedded Systems
- AI and Automation
- Wireless Communication

PROJECTS

SoilBot AI – Intelligent LoRaWAN-Based Smart Soil Monitoring System — 2025 – 2026

- Developed an IoT-based smart agriculture system monitoring soil moisture, temperature, pH, and NPK nutrients using ESP32 and multiple sensors.
- Implemented long-range wireless communication via SX1262 LoRaWAN modules and The Things Network (TTN) for remote agricultural monitoring.
- Built a full-stack web dashboard (HTML/CSS/JS + Python Flask) with live monitoring, trend analysis, and alert generation — deployed at soil-bot.vercel.app.
- Integrated an AI chatbot for irrigation planning, fertilizer management, and soil health recommendations to support precision farming.

Tech Stack: *ESP32, SX1262, LoRaWAN (TTN), DS18B20, RS485 NPK Sensor, HTML/CSS/JS, Python Flask, ThingSpeak, Vercel*

IoT-Based Smart Parking System — Jan 2026 – Jun 2026

- Built using STM32 Nucleo board and ToF (Time-of-Flight) sensors to calculate real-time distance and detect vehicle presence in parking slots.
- Designed embedded firmware for sensor data acquisition and slot occupancy logic on STM32 platform.

Tech Stack: *STM32 Nucleo, ToF Sensor, STM32CubeIDE, Embedded C*

Public Transport Vehicle Tracking System — 2023 – 2024

- Developed a bus tracking system using LoRa SX1278 master-slave architecture for GPS coordinate transmission.
- Integrated a POS machine for real-time seat availability updates linked to live bus location tracking.

Tech Stack: *Arduino Uno, LoRa SX1278, NEO-6M GPS, TinyGPS++*

Automatic Solar Tracking System — 2024

- Designed a solar panel system that auto-adjusts orientation using LDR sensors and servo motors to maximize energy capture throughout the day.

Tech Stack: *Arduino Uno, LDR sensors, Servo motors, Arduino IDE*

CERTIFICATIONS

- Python for Data Science (2025)
- Machine Learning Onramp – MATLAB Online Certification (2024)
- Learning Python for Data Analysis and Visualization – Udemy (2022)
- Introduction to Structured Query Language (SQL) – Coursera (2022)
- The Bits and Bytes of Computer Networking – Coursera (2021)

PAPER PRESENTATION

ICREACT-2023 – International Conference on Research in Electronics Engineering and Communication Techniques, Chennai.